

TABLE 5—EFFECTS OF CAPS ON MORTALITY OUTCOMES: DEATHS PER 10,000 PEOPLE

	2-year effect (1)	5-year effect (2)	Placebo effect (3)	Mean at baseline (4)
Deaths of despair	−0.006 (0.034)	−0.022 (0.032)	−0.007 (0.010)	2.038
Suicide	0.020 (0.018)	0.002 (0.017)	−0.008 (0.005)	0.512
Overdose	0.002 (0.003)	0.001 (0.003)	−0.001 (0.001)	0.015
Alcohol related	−0.024 (0.029)	−0.018 (0.027)	−0.001 (0.012)	1.349
Homicides	0.082 (0.041)	0.149 (0.044)	0.016 (0.012)	1.940
Infant mortality	0.183 (6.341)	−0.001 (2.445)	−0.013 (0.79)	173.782
All-cause mortality	−0.040 (0.174)	0.042 (0.170)	0.035 (0.063)	47.104

Notes: This table reports the average effects of CAPS on the number of deaths related to mental health per 10,000 people. Deaths of despair is an aggregate category that includes deaths by suicide (ICD-10 X60-X84), overdose (ICD-10 X40-X45, Y10-Y15, Y45, Y47, Y49), and alcohol-related deaths (which are a combination of alcoholic liver disease (ICD-10 K70, K73-K74) and deaths by mental disorders, which in the data are mainly associated with alcohol (ICD-10 F00-F99)). Standard errors in parentheses are computed using a municipality-level clustered bootstrap. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. Columns 1 and 2 report the average effect after two years and after five years, respectively. Column 3 reports the placebo effect. Column 4 shows the mean of each variable at baseline—i.e., one period before the event.

trends in homicide rates. The creation of CAPS is associated with a significant increase of 0.149 (s.e., 0.04) deaths per 10,000 people on average, or 7.7 percent of the value at baseline.¹⁹

The Penrose hypothesis is a particularly prominent and extensively discussed explanation for our results. In particular, psychiatric hospitals have an incapacitation effect on mentally ill individuals who may be prone to violent incidents (as perpetrators or victims). Thus, substituting psychiatric hospitals with CAPS could increase rates of violent crimes, especially considering that most of the substitution has affected patients with schizophrenia. In principle, other channels can be consistent with our results. For instance, the increased presence of psychiatric patients in the community can make the neighborhood feel unsafe and increase crime, or CAPS may make it easier for criminals to find victims. Although we cannot directly rule out these other potential explanations, we focus on the Penrose hypothesis. Next, we conduct a quantitative exercise to investigate the plausibility of our results under this mechanism.

¹⁹ Even though the numbers seem large, when put in perspective to externalities from other Brazilian policies on homicides—such as the trade liberalization (Dix-Carneiro, Soares, and Ulyssea 2018) and the transition of a market from legal to illegal (Chimeli and Soares 2017)—our results have much lower magnitude.

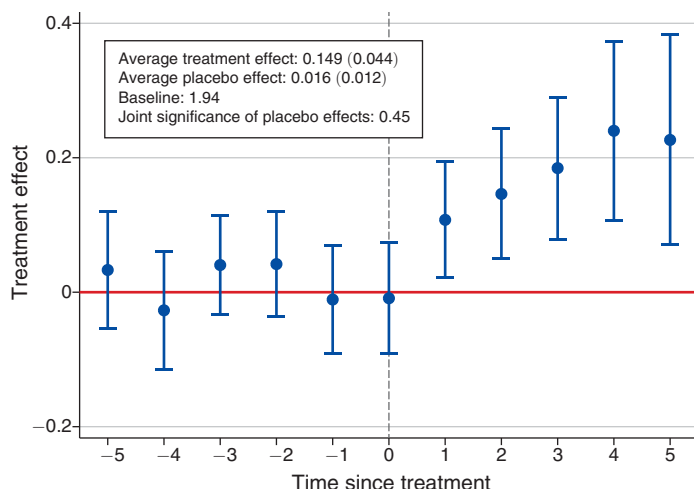


FIGURE 8. EFFECTS OF CAPS ON HOMICIDES: DEATHS PER 10,000 PEOPLE

Notes: This figure plots 95 percent confidence intervals computed with a municipality-level clustered bootstrap and dynamic and placebo DID estimators for the CAPS' effects on the number of deaths per 10,000 people by homicide (ICD-10 X85-Y09). Placebo DID estimators estimate the CAPS' effects had the treatment occurred in a placebo, pretreatment period. They use a varying baseline period—the one immediately before the placebo treatment date—so we do not normalize relative to a unique period. Placebo DID estimators estimate the CAPS' effects had the treatment occurred in a placebo, pretreatment period. They use a varying baseline period—the one immediately before the placebo treatment date—so we do not normalize relative to a unique period. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. The Average Treatment Effect computes a weighted average of the dynamic estimators, giving to each estimator a weight proportional to the number of switchers the estimator applies to. In parentheses, standard errors are computed with a municipality-level clustered bootstrap. Baseline indicates the sample mean values for the treated in the first pre-CAPS period.

If the only channel by which CAPS affect mortality by assault is through dehospitalization, the ratio between the CAPS' effects on homicides and the CAPS' effects on mental health hospitalizations estimates the impact of deinstitutionalization on homicides induced by the policy reform. As the average effect of CAPS on psychiatric admissions is -0.95 , our reduced-form estimates predict that 15.7 percent of deinstitutionalized patients are involved in a violent death. Based on medical research that follows mentally ill patients years after a psychiatric discharge, it is plausible to use as a benchmark that at least 40 percent of discharged inpatients eventually get involved in a violent crime, either as victims (Hiday et al. 1999; Walsh et al. 2003) or perpetrators (Link, Andrews, and Cullen 1992; Fazel et al. 2009; Fleischman et al. 2014).

Even if our reduced-form prediction is within this range, one could argue that violent crime is a broader category, and homicides represent just a small fraction. Still, they are overrepresented among the severely mentally ill. Indeed, Fazel and Grann (2006) find that the population-attributable risk of patients with severe mental illness concerning homicides is 18 percent, the highest among all types of violent crime (it

is 5 percent for all violent crimes). Moreover, data from Murray, de Castro Cerqueira, and Kahn (2013) imply that roughly 1 in 15 violent crimes in Brazil is a homicide. We can then perform a back-of-the-envelope calculation to estimate the ratio of the number of homicides in which severely mentally ill people are involved to the number of violent crimes involving a mentally ill agent by calculating

$$r = \frac{0.18 \times 1}{0.05 \times 15} = 0.24$$

Multiplying this number by the benchmark that 40 percent of discharged inpatients are eventually involved in a violent crime, we calculate that 9.6 percent of discharged patients might become involved in homicides.²⁰ Therefore, if we take our point estimates literally, this exercise illustrates that the effect size we find is within a plausible range. However, given the extent of the noise in the estimation, we remain cautious in our interpretation. Depending on whether one increases or decreases the CAPS' effects on homicides and hospitalization rates by one standard error, the reduced-form prediction varies from 8 percent to 28 percent, making the channel we propose more or less likely.

Our results are consistent with the hypothesis from some criminologists that inpatient mental care can affect crime via an incapacitation effect (Lamb 2015)—i.e., that being treated in an inpatient setting prevents these patients from committing or becoming victims of violent crimes. This is particularly relevant in our context, given that most people switching away from inpatient care are patients with schizophrenia, who tend to be more exposed to violent incidents than the general population (Brekke et al. 2001a; Fazel et al. 2014; Fleischman et al. 2014). However, given the data limitations, we cannot explore this channel further.

V. Robustness Checks

A. Unobserved Policy Changes

One concern regarding the effects we find is that the presence of CAPS could be related to unobserved policy changes that might also affect our main outcomes. Indeed, during the 2000s, Brazil saw other important public policies, like the PBF and Family Health Program (*Programa Saúde da Família*—henceforth, PSF), created or expanded substantially. To partially address these concerns, we control for differential trends according to PBF spending changes and state-year fixed effects, which account for contemporaneous programs partially determined at the state level. Still, one might worry whether the timing of CAPS adoption correlates with the rollout of PSF across municipalities, which is known to improve health outcomes.

We believe this should be less of a concern, given that PSF municipality coverage had already reached nearly 80 percent when psychiatric reform began. What is more, the correlation between the timing of these programs is low (see online

²⁰ We thank an anonymous referee for suggesting this discussion.

Appendix Figure A8 for a scatter plot).²¹ Still, we test the stability of our baseline estimates to the inclusion of controls for PSF adoption and adoption year $\times$ time trends (see online Appendix Table A6).²² Our results are unsurprisingly robust to these alternative specifications. Finally, given previous research documenting remarkable negative effects of PSF on infant mortality (Bhalotra, Rocha, and Soares 2019; Macinko et al. 2007), and the fact that other Brazilian health care policies have significantly affected inputs in the production function of infant health (Carrillo and Feres 2019), we assess the impact of CAPS on infant mortality and interpret it as a placebo test. Consistent with previous results, online Appendix Figure A9, panel A shows that the effects are flat around zero. We also estimate effects flat around zero for all-cause mortality (panel B). The evidence above is consistent with the interpretation that our results capture the effects of CAPS and not of other unobserved policy changes.

B. *Alternative Specifications*

In principle, our strategy should be independent of including controls. Finding that results change significantly depending on the controls used could threaten our identifying assumption. In columns 2 and 3 of online Appendix Table A7, we report the average effects on our main outcomes varying the set of controls used in the estimation. In column 2, we estimate the effects without any control. In column 3, we present the estimates from a specification without controlling for our time-varying controls: population composition by age and gender, GDP per capita, and PBF expenditures per capita. Results remain similar, with slightly larger but not substantial differences in the specification without controls. The same table reports our main results when we weigh municipalities by baseline population (column 4). The standard errors increase remarkably in this specification, so all our results remain within reasonable bounds.²³

Column 5 reports the average effect for our main outcomes using a restricted, treated sample that includes only municipalities with at least five years of exposure to treatment. All our main results remain stable when we explore this alternative composition of treated cohorts. These results indicate that compositional changes (caused by the fact that some late-treated cities have missing post-CAPS data) do not affect our results. There also exists a possibility that our choice of restricting the event time to the fifth year after CAPS adoption trims an important part of the effects if they take more time to fully materialize. To assess whether our results significantly underestimate the effects, column 6 reports the average effects for our main outcomes, extending the event time horizon to eight years after CAPS. Indeed, we see slightly stronger effects. However, we estimate that the effects are already stable

²¹ The raw correlation is 0.06. If we weight observations based on baseline population, it decreases to -0.03, indicating that larger municipalities are not more likely to adopt both programs in parallel.

²² We thank Bhalotra, Rocha, and Soares (2019) for providing data on the implementation of PSF across municipalities.

²³ As shown by Solon, Haider, and Wooldridge (2015), if the assumed covariance structure is a good approximation to the true one but not exactly the right one, including weights leads to less precise estimates even in the absence of heterogeneity in treatment effects.